

Snippet Integration Test

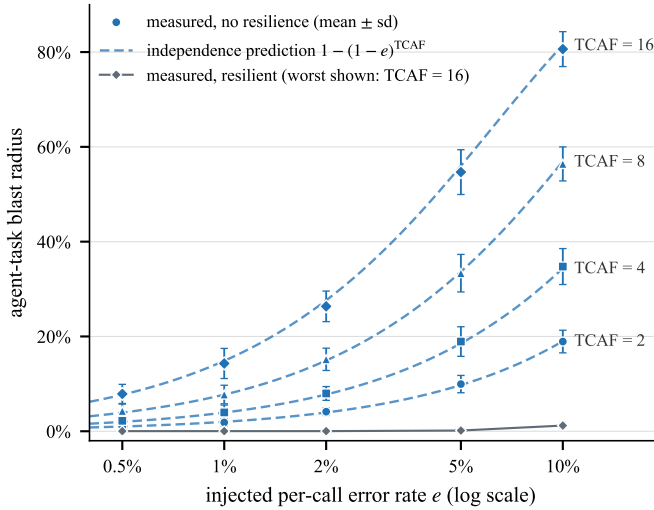


Fig. 1. Measured agent-task blast radius versus injected per-call error rate e (log scale), 20 seeds $\times$ 200 tasks per cell (the Table I sweep). Without resilience (blue, mean $\pm$ sd) blast radius tracks the independence prediction $1 - (1 - e)^{\text{TCAF}}$ (dashed) across the full grid; the §VII resilience patterns (gray; worst case, TCAF = 16, shown) recover it to $\leq 1.2\%$ everywhere. Generated by paper/figures/ in the released harness.

TABLE I

Measured blast radius (%) under correlated faults at the same $\sim 2\%$ marginal per-call error rate, TCAF = 8 (independence prediction: 14.9%); mean $\pm$ sd over 50 seeds $\times$ 200 tasks. Lock-in: seeds whose resilient blast radius exceeds 50% — a transient fault trips the ETA breaker, which has no half-open path in the Table I configuration. +probes: same patterns with a breaker probe every 8 skipped calls.

condition	marg. e	BR, no resil.	BR, resilient	lock- in	BR, +probes
iid	2.0%	15.1 ± 2.6	0.0 ± 0.1	0/50	0.0 ± 0.1
bursts, mean 4	2.3%	6.0 ± 2.2	2.1 ± 1.0	0/50	2.1 ± 1.0
bursts, mean 16	2.5%	3.5 ± 2.7	5.3 ± 19.5	2/50	1.6 ± 1.9
bursts, mean 64	2.9%	3.2 ± 5.0	3.0 ± 14.1	1/50	1.4 ± 3.3
server episodes (20)	2.4%	9.8 ± 11.9	20.8 ± 33.5	8/50	10.1 ± 12.2

I. Blast-radius figure (for §VI)

Body filler to exercise column layout. The tool-call plane compounds per-call failure into task failure through fan-out; the figure shows the measured sweep against the closed-form prediction. This paragraph exists only to give the floats somewhere to sit and to surface overfull boxes at real column width.

II. Correlated-fault table (for §VI)

More filler so the table float has an anchor paragraph and the two-column balance is realistic. Correlation concentrates failure into heavy tails, defeats retry budgets, and exposes breaker lock-in.

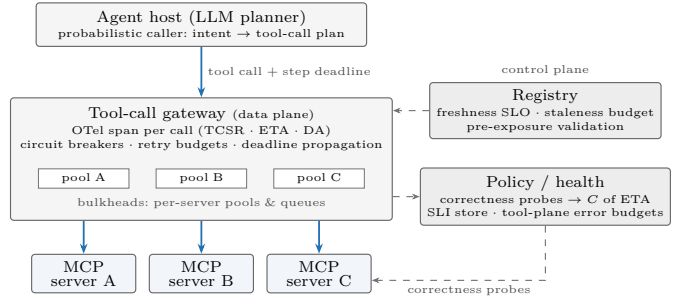


Fig. 2. The managed tool-call plane (§VIII). Solid blue — data plane: every call crosses a gateway that emits an OpenTelemetry span carrying TCSR/ETA/DA attributes, enforces circuit breakers and retry budgets, propagates deadlines, and isolates servers behind per-server bulkhead pools. Dashed gray — control plane: a registry with a freshness SLO performs pre-exposure validation and feeds the gateway its validated tool set; a policy/health layer ingests the gateway’s spans/SLIs and runs correctness probes (the C factor of ETA) against servers. Deliberately analogous to a service-mesh data plane plus control plane.

III. Reference architecture (for §VIII)

Final filler paragraph. The architecture is the operational complement of the SLIs: the gateway is where TCSR, ETA, and DA are emitted and enforced, and the control plane is where tool-plane error budgets are spent.